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1. Write a program to implement Floyd Warshall Algorithm to find the shortest path between every pair of the vertices.

Program:

```
#include<stdio.h>
#define INF 99999
int main(){
int n,i,j,k;
int a[10][10];
printf("Enter number of vertices: ");
scanf("%d",&n);
printf("Enter the adjacency matrix (use 99999 for INF):\n");
for(i=0;i<n;i++){
for(j=0;j<n;j++){
scanf("%d",&a[i][j]);
}
}
for(k=0;k<n;k++){
for(i=0;i<n;i++){
for(j=0;j<n;j++){
if(a[i][k]!=INF&&a[k][j]!=INF&&a[i][k]+a[k][j]<a[i][j]){
a[i][j]=a[i][k]+a[k][j];
}
}
}
}
printf("\nShortest Distance Matrix:\n");
for(i=0;i<n;i++){
for(j=0;j<n;j++){
if(a[i][j]==INF)
printf("INF\t");
else
printf("%d\t",a[i][j]);
}
printf("\n");
```

```
}
return 0;
}
```

Terminal Process:

